
ANALYSIS OF CNN PERFORMANCE FOR MONKEYPOX CLASSIFICATION USING TRANSFER LEARNING

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ABSTRACT

Monkeypox is a viral zoonotic disease passed from animals to humans, part of the Orthopoxvirus group, causing flu-like signs including fever, skin sores, or enlarged glands. Although it usually goes away on its own, rising infection numbers show serious risks for weaker groups - underlining gaps in detection methods and public understanding. Here, we suggest building a precise deep-learning system able to quickly identify monkeypox using pictures of skin lesions. Leveraging Convolutional Neural Networks (CNNs), the methodology utilizes the "Monkeypox Skin Image Dataset" to automate the detection process. The suggested model analyzes images using stacked convolutional and pooling steps, then applies a dense layer to sort skin conditions into three types: Monkeypox, Normal, or Other.

Keywords Monkeypox · Convolutional Neural Networks (CNN) · Deep Learning, Skin Lesion Classification · Orthopoxvirus · Automated Detection

1 Introduction

A Monkeypox comes from a virus related to smallpox, found in the Orthopoxvirus group; it spreads when people touch sick animals or others who are infected. It shows symptoms like fever, skin sores, and tender glands - unlike smallpox which doesn't affect glands this way - but usually goes away on its own over time. Still, some folks, especially those with weak immune systems, might get very ill. Lately, more cases have popped up around the world, showing we really need better ways to catch the infection early without delay. In 1958, sick monkeys that had been sent from Singapore to a research center in Denmark provided the initial clues that led to the isolation and identification of the monkeypox virus (Kumar, Acharya, Gendelman, & Byrareddy, 2022).

Folks often use PCR tests to spot monkeypox, but that needs labs, skilled workers - also takes a while. When big outbreaks hit - or clinics lack supplies - that's when things get tricky. Still, more skin sore pics are now online; paired with better image software - it's becoming easier to catch diseases using smart tools.

Deep learning - especially methods like CNNs - has become key for sorting medical images. These networks grab important details straight from photos, so they spot tiny clues in skin issues people might miss. New research shows they work well spotting illnesses including melanoma, chickenpox, or even rare ones like monkeypox.

This project works on creating a tool that spots monkeypox through skin images, using CNN models trained on data from Kaggle. Instead of manual checks, it uses smart algorithms to tell apart monkeypox rashes from similar-looking issues. With this approach, the goal is faster identification - helping doctors act sooner. Because it runs on open datasets, others can test or expand it later. The method could ease pressure on clinics during outbreaks while boosting response times in communities.

2 Related Work

The jump in monkeypox infections lately pushed scientists to build faster, easier-to-use tests - helping people understand risks better while slowing transmission. Instead of waiting days for results based on signs like fever or skin sores, doctors now face delays and steep bills using old-school checkups. Because of that, more studies have started leaning on machine learning models - to speed up detection without relying only on physical exams.

The way we diagnose diseases is quickly shifting to deep learning, especially with tools called CNNs. Since these systems handle pixel-based info well, they work great for checking medical scans. Even though there aren't many big monkeypox image sets around, new research shows it's possible to spot skin lesions automatically.

One key study helped shape the approach - another played a big role too

1. Robust Deep Convolutional Neural Network (Bala, 2022)

Diponkor Bala looked into better ways to sort out diseases correctly. His idea? A strong neural network built just for spotting monkeypox. Instead of general models, he focused on one that learns patterns in skin images deeply. Because of this, machines might now tell monkeypox apart from similar rashes more reliably. That early result gives others a starting point - something real to build on down the line.

2. Monkeypox Diagnostic-Aid System (Muñoz-Saavedra et al., 2022)

Building from how CNNs are used, Muñoz-Saavedra with Civit-Masot, Luna-Perejón, Civit, and Domínguez-Morales built a system meant to help spot monkeypox. Instead of just theory, they trained a network using pictures of skin conditions. The goal wasn't full automation but support - giving doctors an extra hand when making calls. By leaning into real-world use, their approach highlights tools that can catch illness sooner, especially where people face higher risk.

3. Monkeypox Skin Lesion Detection (Ali et al., 2022)

In separate attempts to tackle missing data issues, Ali's team ran an early test using the "Monkeypox Skin Lesion Dataset" (MSLD). Instead of building models from scratch, they tested ready-made ones - like VGG16 and ResNet50 - to see how well these could tell monkeypox apart from illnesses such as chickenpox or measles. The findings matter because they show that borrowing knowledge through Transfer Learning helps when there isn't much data available - an issue this project also runs into.

4. Deep Learning-Based Model using Modified VGG16 (Ahsan et al., 2022)

Looking deeper into how CNNs work, Ahsan's team tweaked the VGG16 setup to spot monkeypox signs. Instead of just using raw images, they boosted training variety through smart reshaping methods - this helped the system adapt better. When stacked up against classic models, their adjusted version showed stronger results, mainly by updating only key layers. This hands-on approach proves tailored networks might outperform one-size-fits-all designs when checking skin conditions

3 Methods

3.1 Data Set

Our dataset consists of 769 RGB images representing human skin conditions, specifically selected to detect **Monkeypox**. There are three distinct classes in our dataset: **Monkeypox**, **Normal**, and **Others**.

For the Monkeypox class, we have 278 images. For the Normal and Others classes, we have 293 and 198 photos, respectively. We collected our dataset from 2 sources and customised the class labels, grouping all skin diseases (except Monkeypox) such as chickenpox and measles, into other classes, so our model can distinguish Monkeypox from other skin diseases and normal skin.

Data Preprocessing: To prepare the data for the VGG16 architecture, we resized all images to 224 x 224 pixels and normalized them by scaling pixel intensity values to the range [0, 1].

Data Augmentation: We use data augmentation to prevent overfitting because the dataset is small. We applied ImageDataGenerator for random transformations during training, including a rotation range of 20 degrees, a zoom range of 15%, a width/height scaling factor of 0.2, and horizontal flipping.

Data Distribution and Class Imbalance: The dataset comprises a total of approximately 600 training samples and 150 testing samples. A critical analysis of the class distribution shows a significant imbalance.

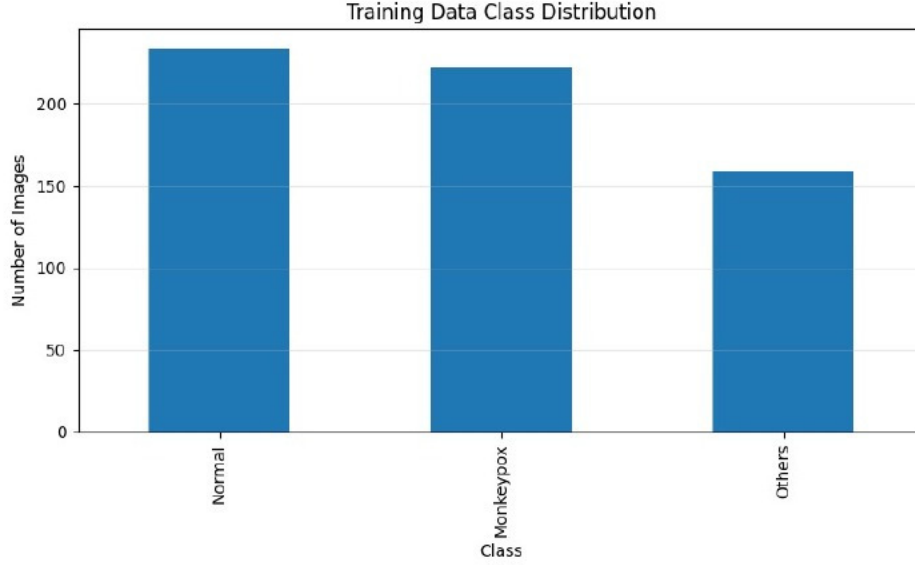


Figure 1: Histogram of Training Dataset

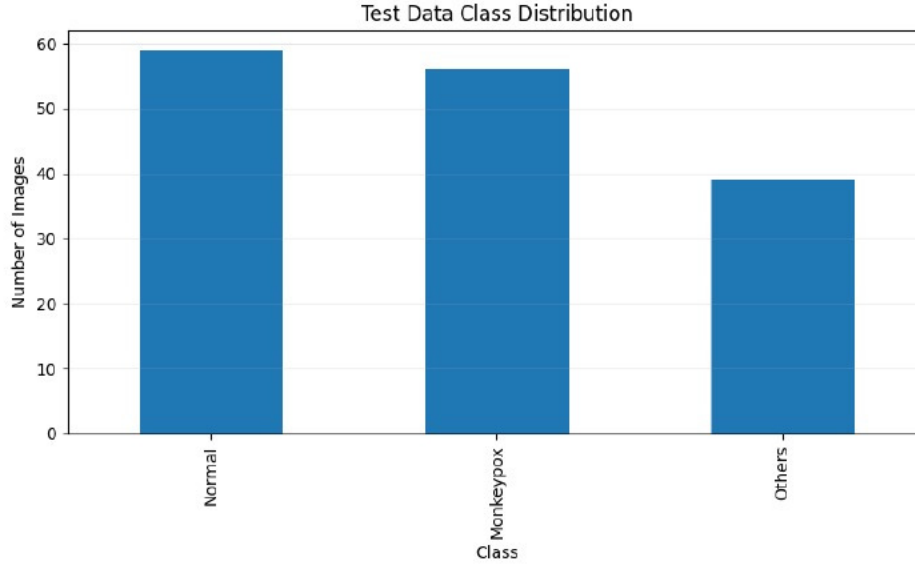


Figure 2: Histogram Of Test Dataset

Figure 1 :shows the distribution of the three classes. Data imbalance is noticeable here for the Normal class. The Monkeypox class has the fewest images, while the Normal class has the most, which can pose challenges for training an accurate classification model. Addressing this imbalance is essential for ensuring reliable results.

Figure 2: shows the distribution of the test set. The test dataset also illustrates the same ratio as the train dataset. This indicates a possibility that evaluation metrics reflect the model's performance across all categories fairly.

3.2 Network Architecture

First, we trained a base CNN model on our dataset. However, due to overfitting, it did not perform well and took a long time to train.

Our final deep learning framework is based on transfer learning to address the problem of distinguishing Monkeypox from visually similar dermatological diseases using a small dataset. We used the VGG16 architecture as our feature

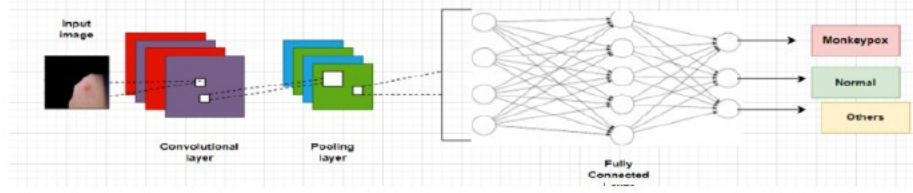


Figure 3: Base CNN Model

backbone rather than training a deep network from scratch, which frequently results in overfitting on tiny datasets. VGG16 convolutional model was already pre-trained on the massive ImageNet dataset. It consists of several blocks with small 3×3 filters. These layers work as hierarchical feature extractors. The early layers detect simple details like edges, textures, and colors. The deeper layers recognize complex patterns, such as the shapes of skin lesions. By freezing these pre-trained weights, we successfully used this broad visual knowledge for our specific task of classifying skin diseases. One of the most important parts of our design was adapting the classification 'head' using the GAP strategy. Traditional VGG16 models usually contain a 'Flatten' layer connected to enormous fully connected layers. This creates an enormous number of parameters. Whenever the amount of training data is small, this high number of parameters increases the chances of overfitting, i.e., the model memorizes instead of learning. To mitigate this, we replaced the Flatten operation with **Global Average Pooling (GAP)**.

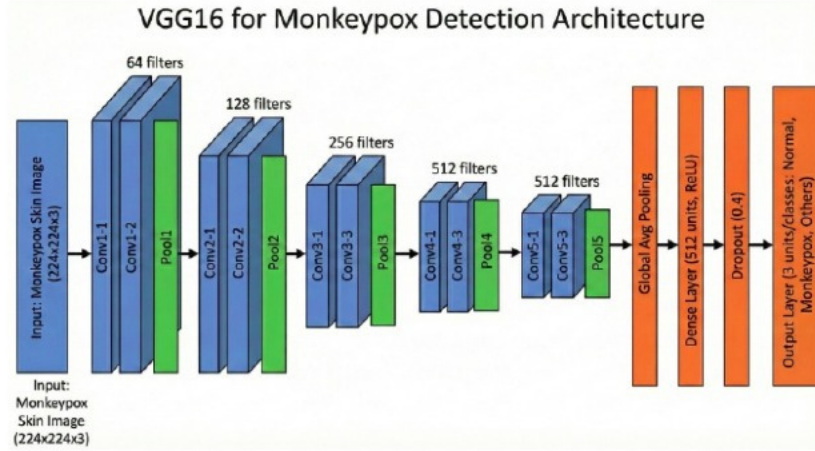


Figure 4: Network Architecture of VGG16

The input image is processed by the frozen VGG16 encoder. The extracted features are then passed through a Global Average Pooling (GAP) layer, followed by a Dense layer with Dropout regularization, and finally a Softmax activation layer (multiclass classification) output layer.

Configuration of layers:

1. **Global Average Pooling:** Condenses the spatial feature maps ($7 \times 7 \times 512$) into a 1D vector (1×512), drastically reducing the number of parameters.
2. **Dense Layer:** 512 neurons activated by the ReLU function to interpret the extracted features.
3. **Dropout:** We applied a 0.4 dropout rate to randomly deactivate 40% of the neurons during training, preventing co-adaptation.
4. **Output:** A **Softmax** layer with three neurons providing probabilities for Monkeypox, Normal, and Others, as we have more than 2 classes.

3.3 Experiments

We conducted a comprehensive set of experiments to determine the most effective model configuration. To support the architectural decisions, the experimental phase concentrated on comparing our Transfer Learning approach with a proprietary baseline model.

Model Selection: Custom Base CNN vs. VGG16 To establish a performance benchmark, we first designed and trained a custom Convolutional Neural Network (Base CNN) from scratch.

Base CNN Setup: The baseline model consisted of 3 convolutional blocks utilizing AveragePooling2D and BatchNormalization. The head utilized a traditional Flatten layer followed by a Dense layer of 512 units.

The "Parameter Explosion" Problem:: A critical failure mode observed in the Base CNN was the dimensionality explosion caused by the Flatten layer. As recorded in our experimental logs, this architecture resulted in a staggering 51,476,419 trainable parameters.

Outcome: Despite the high complexity, the Custom Base CNN struggled to generalize on the limited dataset, achieving a test accuracy of only 77.27%.

Our Transfer Learning-based VGG16 architecture showed superior effectiveness. We decreased the trainable parameters by 99.5% (to about 264k) by using pre-learned filters and substituting Global Average Pooling (GAP) for the Flatten layer. The model was able to attain a much higher test accuracy of about 88% through this lightweight approach.

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Table 1: Performance Comparison Summary

Metric	Custom Base CNN	VGG16
Approach	Trained from Scratch	Transfer Learning (Frozen)
Pooling Type	Average Pooling	Global Average Pooling (GAP)
Total Parameters	51,476,419 (Heavy)	264,195 (Lightweight)
Test Accuracy	77.27%	~88.00%
Observation	Overfitting due to complexity	Generalized well

Training Configuration and Regularization To optimize the learning process and handle class imbalance, we implemented the following strategies: **Optimizer:** We selected the Adam optimizer initialized with a learning rate of 0.0005, which outperformed Stochastic Gradient Descent (SGD) in terms of convergence speed on this loss landscape.

Callbacks: We employed EarlyStopping (patience=8) to prevent over-training and ReduceLROnPlateau (patience=3, factor=0.5) to fine-tune the learning rate dynamically when the validation loss plateaued.

Training Dynamics and Results Our ultimate model was trained for 30 epochs with a batch size of 32.

Figure 5(a): Training and test accuracy over 30 epochs. The training accuracy (blue) reaches ~98%, indicating strong feature learning, while test accuracy stabilizes around 88%.

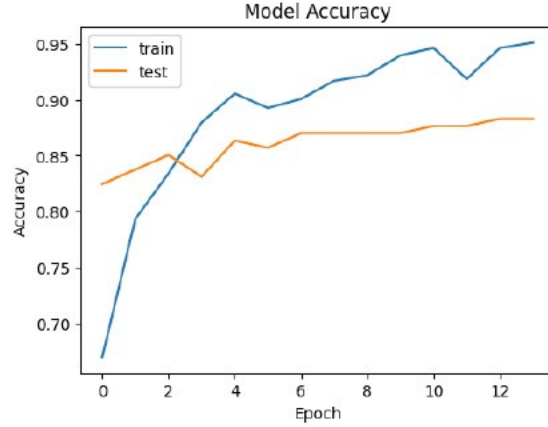


Figure 5: Model Accuracy

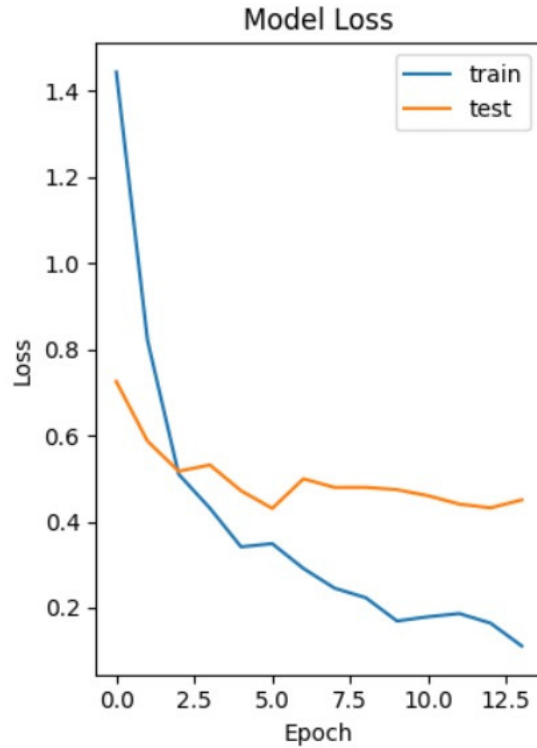


Figure 6: Model Loss

Figure 6: Training and test loss over 30 epochs. The test loss (orange) settling in around 0.5 suggests the limit of generalization has been reached, given the dataset size.

4 Results

4.1 Experimental Setup Overview

The proposed Convolutional Neural Network (CNN) was trained on a dataset of 769 images classified into three categories: Monkeypox, Normal, and Others. The dataset was split using a stratified hold-out method where 80% was allocated for training and 20% for testing. Data augmentation features such as rotation, zoom, shear, and flips were applied to the training set to improve the model's generalization. The model was trained using the Adam

optimizer (initial learning rate = 0.0005) with categorical cross-entropy loss. Callbacks for Early Stopping and ReduceLROnPlateau (a learning rate scheduler) were utilized to optimize training efficiency.

To address the limitations observed in the initial Custom Base CNN model (specifically the low performance on the "Others" class), we transitioned to a Transfer Learning approach and used VGG16 for comparison, aiming for greater accuracy and precision. The dataset (N=769) was split into 80% training and 20% testing using a stratified sampling strategy to ensure consistent class distribution.

4.2 Classification Performance

Upon evaluating the model on the unseen test set (N=154), the base CNN model achieved a Final Test Accuracy of 79.22% and a Test Loss of 0.6684. The training process terminated at Epoch 26 due to the Early Stopping mechanism, indicating that the validation loss had plateaued.

When we tested the new VGG16 Transfer Learning model on the 154 images it hadn't seen before, the performance was excellent. It achieved a final test accuracy of 85.71% and managed to keep the prediction error (Test Loss) really low at just 0.4306. The training was efficient; the model converged and found its peak performance quickly, finishing its run successfully after only around 14 epochs.

Table 2: Classification Report by Class (Custom Base CNN)

Class	Precision	Recall	F1-Score	Support
Monkeypox	0.80	0.82	0.81	56
Normal	0.87	0.83	0.85	60
Others	0.65	0.68	0.66	38
Weighted Avg	0.79	0.79	0.79	154

Table 3: Classification Report by Class (VGG16 Transfer Learning)

Class	Precision	Recall	F1-Score	Support
Monkeypox	0.87	0.82	0.84	56
Normal	0.92	0.92	0.92	59
Others	0.76	0.82	0.79	39
Weighted Avg	0.86	0.86	0.86	154

As shown in Table 2 the model performed most actively on the Normal class, achieving the highest F1-score of 0.83. The Monkeypox class also performed well, with a recall of 0.82. However, performance dropped significantly for the Others class (F1-score of 0.65), which acted as the limiting factor for overall accuracy.

On the other hand, Table 3 shows that the VGG16 Transfer Learning model provides an overall improvement over the Custom Base CNN. The Normal class remained the most robustly classified category, achieving a nearly perfect F1 Score of 0.92. The Monkeypox class continued to show high performance, with its Precision increasing to 0.87. Crucially, the performance for the problematic Others class saw a dramatic jump, with the F1-score rising significantly to 0.79 (up from 0.66), driven by a significant improvement in recall to 0.82 (up from 0.68). This indicates that the VGG16 model largely overcame the core limitation of the previous architecture, leading to a much more balanced classifier.

Consequently, the Weighted Average F1-score for the entire model rose to 0.86, reflecting the overall improvement in the VGG16 Transfer Learning approach.

Overall Improvement:

- Custom Base CNN: 79% weighted average across all metrics
- VGG16 Transfer Learning: 86% weighted average
- Gain: +7 percentage points (8.9% relative improvement)

1.Win: The "Others" class improved by 13 percentage points (66% to 79%) with transfer learning, addressing the weakest area of the Custom CNN.

2.Consistent Gains: VGG16 outperforms Custom Base CNN across all metrics for all classes.

3.Normal Class Excellence: VGG16 achieves 92% across all metrics for Normal images.

4.Clinical Relevance: Monkeypox recall remained at 82%, but precision increased from 80% to 87%, reducing false positives.

4.3 Visual Explanations by Grad-CAM

To verify that the high quantitative performance (85.71% accuracy) translates to clinically relevant features, we performed a visual analysis using Gradient-weighted Class Activation Mapping (Grad-CAM). We extracted the top 10 most confidently classified images for each class (Monkeypox, Normal, Others) to inspect the model's focal points. These visual results, along with an analysis of the most confident misclassifications, are detailed in the Discussion section and for visualization of Prediction of Monkeypox Using Grad Cam Shown in Figure 7.

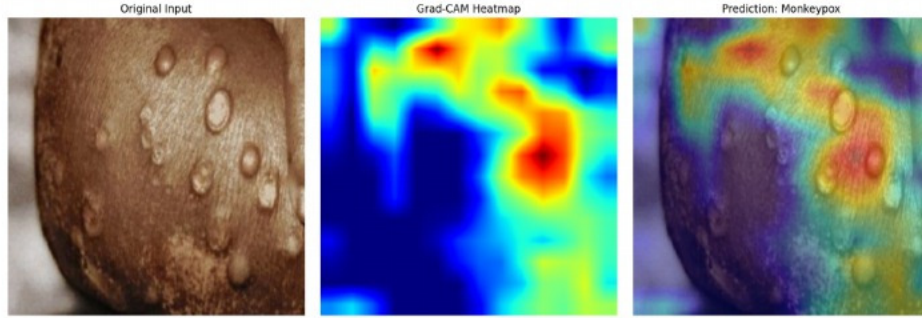


Figure 7: Prediction Of Monkeypox Using Grad CAM

Model Performance Comparison: In Figure 8, key performance insights between Custom Base CNN vs VGG16 .



Figure 8: Custom Base CNN VS VGG16 Model Metrics Comparison

5 Discussion

5.1 Interpretability Analysis using Grad-CAM

To validate the clinical relevance of our VGG16 Transfer Learning model, we used Grad-CAM in this study. This method shows the parts of the input image that the model considers most important for its prediction.

- **Normal Class:** As observed in the top accurate visualizations, the model focuses on broad, uniform areas of skin texture. The heat maps show distributed activation across healthy skin, confirming that the model creates a "baseline" for normality rather than focusing on noise or background artifacts.

- **Monkeypox Class:** The Grad-CAM heatmaps used for correctly classified Monkeypox cases demonstrate that the model successfully localizes the characteristic lesions. Activation peaks (red areas) are usually located above the center of pustules or vesicular rashes. It implies that the model is learning pathologically related features rather than habitual factors (e.g. lighting or background).
- **Others Class:** For non-Monkeypox lesions (Others), the model focuses on the specific irregularities that differentiate them from Monkeypox, such as the different clustering patterns of Chickenpox or the specific texture of Measles.

5.2 Analysis of Prediction Confidence and Errors

To further evaluate the robustness and clinical reliability of the VGG16 model, we conducted a qualitative analysis of the images where the model demonstrated the highest confidence scores, examining both correct predictions and errors.

5.3 Top Accurate Predictions

The model exhibited exceptionally high confidence less than 98 percent, when classifying images that presented clear, unobstructed views of skin lesions with distinct features.

Monkeypox: The most confident and accurate classifications occurred when images clearly depicted the characteristic pustules with central umbilication, a distinctive morphological hallmark of Monkeypox. This suggests that the model has successfully learned to recognize this specific feature, as illustrated in **Figure 9**.

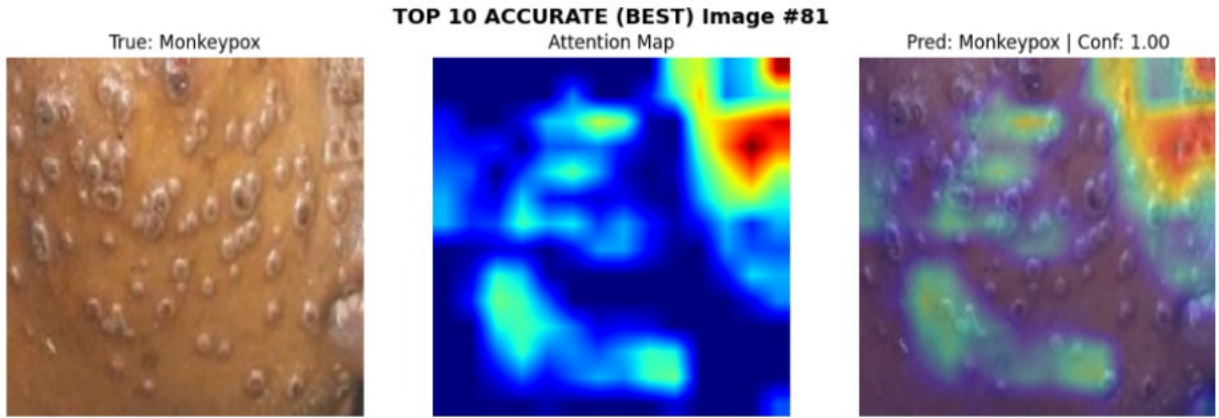


Figure 9: One of the images from Top 10 Accurate images from Monkeypox class

Normal: The model demonstrated near-perfect confidence when analyzing clear patches of healthy skin, confirming it can reliably identify non-pathological tissue without falsely detecting lesions on clean backgrounds, as illustrated in **Figure 10**.

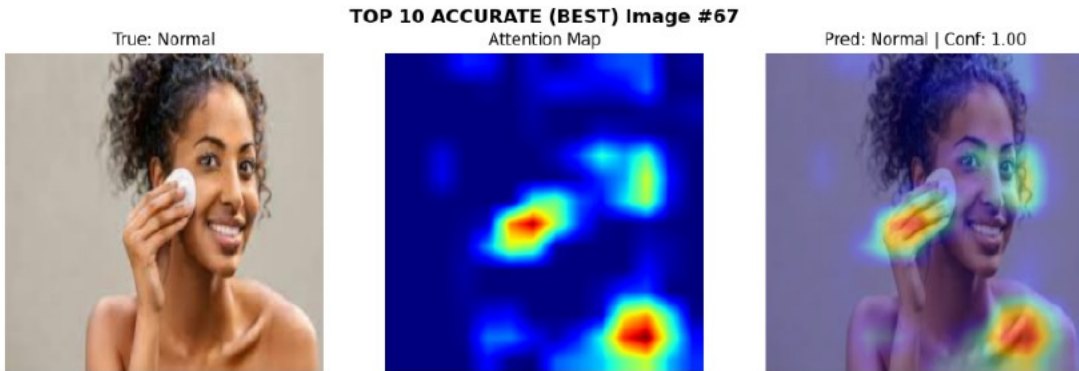


Figure 10: One of the images from Top 10 Accurate images from Normal class

Others: For the "Others" category, the model typically focused on lesions that possessed distinct visual characteristics differentiating them from Monkeypox, such as the varied clustering of Chickenpox blisters or the specific textural appearance of Measles rashes. This indicates the model's capacity to discern specific irregularities associated with other skin conditions, minimizing confusion with Monkeypox. Thats visualized on **Figure 11**.

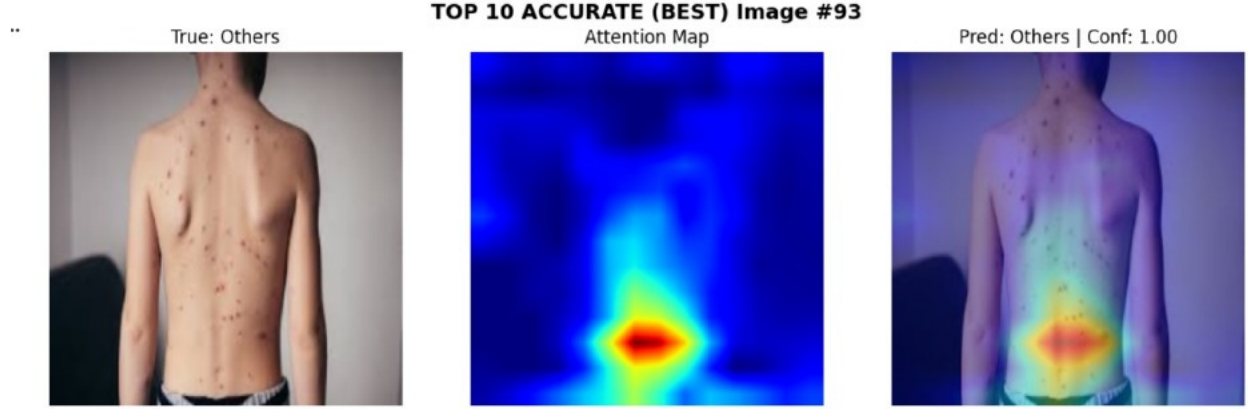


Figure 11: One of the images from Top 10 Accurate images from Others class where chickenPOX detected

6 Misclassifications

Analyzing the top errors revealed specific challenges where the model was confidently incorrect:

Visual Mimicry: The most frequent errors occurred in the "Others" class, particularly with conditions like Chickenpox being misclassified as Monkeypox. These false positives typically involved images of solitary blisters that bear a strong morphological resemblance to early-stage Monkeypox lesions, which is clearly visible on **Figure 12**.

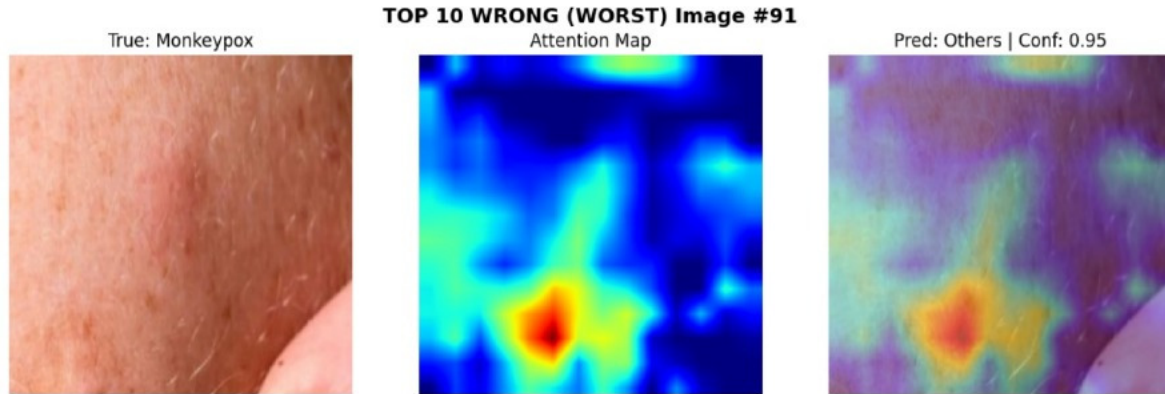


Figure 12: One of the images from Top 10 Wrong images from Normal class

Image Quality Issues: Several misclassified images suffered from poor lighting, blur, or obstruction (e.g., hair or clothing covering the lesion). In these instances, the model likely focused on ambiguous textures rather than definitive pathological features, leading to incorrect classifications.

7 Conclusion

The use of Grad-CAM verifies the claim that the 7% increase in performance while using the VGG16 Model (Weighted F1-Score is 0.86), as compared to the Custom Base CNN, perhaps due to the model learning genuine dermatological features. The remarkable improvement in the "Others" class (F1-score increased from 0.66 to 0.79) is visually confirmed by the heatmaps, as the transfer learning model better tolerates irrelevant skin noise and emphasizes lesion morphology

compared with training from scratch. Using transfer learning with VGG16, we achieved significant performance improvements in classifying skin lesions, especially in the category "Others". Future work will involve further optimizations and deployment of the model for real-world applications.

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